Exp. No: 8 Implementation of Dielectric Free-Space Boundary Conditions in Electric Field and Determination of Relative Permittivity

Aim:

To implement the dielectric–free-space boundary conditions for electric fields and determine the relative permittivity of an unknown medium.

Objectives :

1. To resolve electric-field vectors into normal and tangential components at a dielectric interface.

2. To apply the continuity condition for tangential electric field and the normal flux-density relation.

3. To determine the relative permittivity of an unknown dielectric medium from boundary-field data.

4. To verify the boundary conditions numerically using MATLAB.

Test Case (i) – Normal and Tangential Field Components at the Free-Space / Dielectric Interface

Determine the field components, normal flux density and refraction angle in the dielectric for a specified incident field in free space and a specified relative permittivity εr2.

```
%% Boundary Conditions: Electric Field at a Dielectric Interface

clc;

clear;

close all;


% Physical Constants

eps_0 = 8.854e-12;  % Permittivity of free space (F/m)


%% User Inputs

E1  = input('Enter magnitude of E1 in free space (V/m): ');

th1 = input('Enter angle of E1 with the normal (deg): ');
```

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```
er2 = input('Enter relative permittivity of medium 2: ');


%% Field Calculations in Medium 1
E1_normal = E1 * cosd(th1);
E1_tang   = E1 * sind(th1);
D1_normal = eps_0 * E1_normal;


%% Boundary Conditions & Calculations in Medium 2
% 1. Tangential E is continuous: Et2 = Et1
E2_tang   = E1_tang;


% 2. Normal D is continuous (charge-free boundary, rho_s = 0): Dn2 = Dn1
D2_normal = D1_normal;


% 3. Normal E in medium 2
E2_normal = D2_normal / (eps_0 * er2);


% Magnitude and refraction angle
E2_mag = hypot(E2_normal, E2_tang);
th2    = atand(E2_tang / E2_normal);


%% Console Output (Exact formatting preserved)
fprintf('E1n = %.2f V/m, E1t = %.2f V/m\n', E1_normal, E1_tang);
fprintf('D1n = %.4e C/m^2\n', D1_normal);
fprintf('E2t = %.2f V/m, D2n = %.4e C/m^2\n', E2_tang, D2_normal);
fprintf('E2n = %.2f V/m, |E2| = %.2f V/m\n', E2_normal, E2_mag);
```

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```
fprintf('Refraction angle theta2 = %.2f deg\n', th2);


%% Visualization
figure('Color', 'w');
hold on;


% Interface boundary (x = 0 line)
plot([0, 0], [-80, 80], 'k', 'LineWidth', 2);


% Incident vector (Medium 1, terminating at origin)
quiver(-E1_normal, -E1_tang, E1_normal, E1_tang, 0, 'b', 'LineWidth', 2);


% Refracted vector (Medium 2, originating from origin)
quiver(0, 0, E2_normal, E2_tang, 0, 'r', 'LineWidth', 2);


% Region Labels and Field Annotations
text(-90, 65, 'Medium 1 (free space, \epsilon_0)', 'FontSize', 9);
text(5, -70, sprintf('Medium 2 (dielectric, \\epsilon_{r2} = %.1f)', er2), 'FontSize', 9);
text(-E1_normal * 0.4, -E1_tang * 0.8, ...
    sprintf('E_1 (\\theta_1 = %.0f^\\circ)', th1), 'Color', 'b', 'FontSize', 9);
text(E2_normal * 0.35, E2_tang * 0.8, ...
    sprintf('E_2 (\\theta_2 = %.2f^\\circ)', th2), 'Color', 'r', 'FontSize', 9);


% Plot Styling
axis equal;
```

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```
grid on;

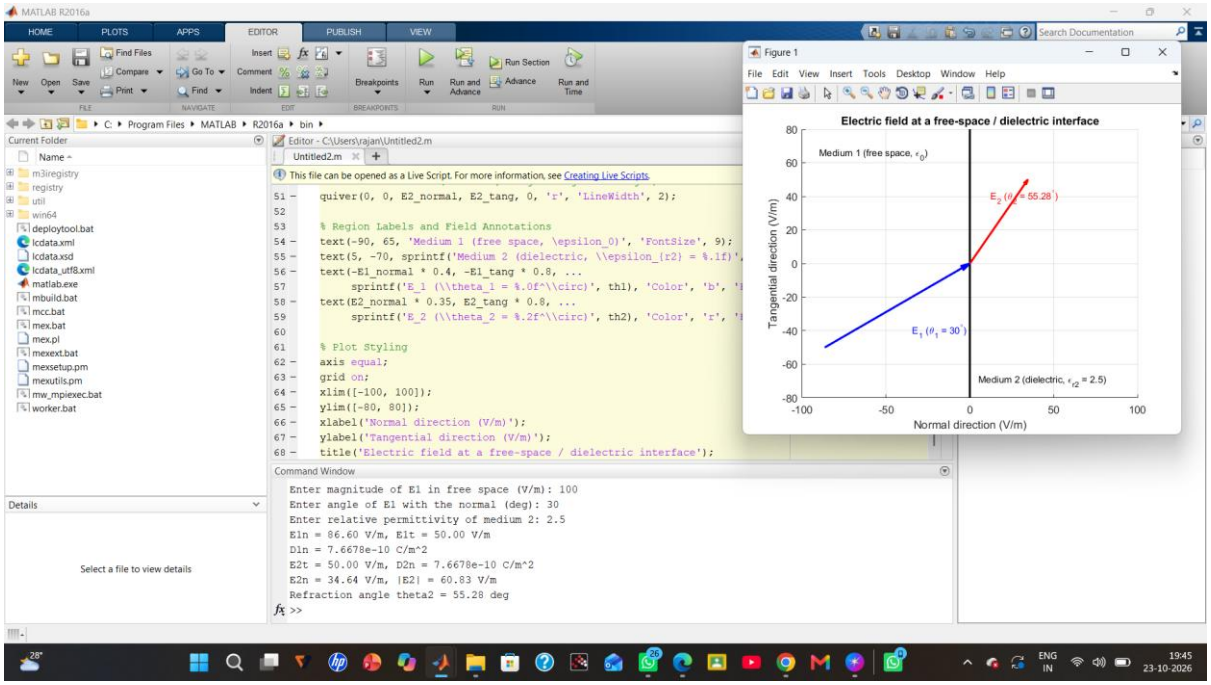
xlim([-100, 100]);

ylim([-80, 80]);

xlabel('Normal direction (V/m)');

ylabel('Tangential direction (V/m)');

title('Electric field at a free-space / dielectric interface');
```



Test Case (ii) – Determination of the Unknown Relative Permittivity from Boundary-Field

Determine the relative permittivity of the dielectric from the normal electric-field components on the two sides of the interface and calculate the percentage deviation from the reference value.

```
clc; clear;

E1n    = input('Enter normal E-field in free space E1n (V/m): ');

E2n    = input('Enter measured normal E-field in the dielectric E2n (V/m): ');

er_ref = input('Enter reference relative permittivity: ');

er_calc = E1n/E2n;

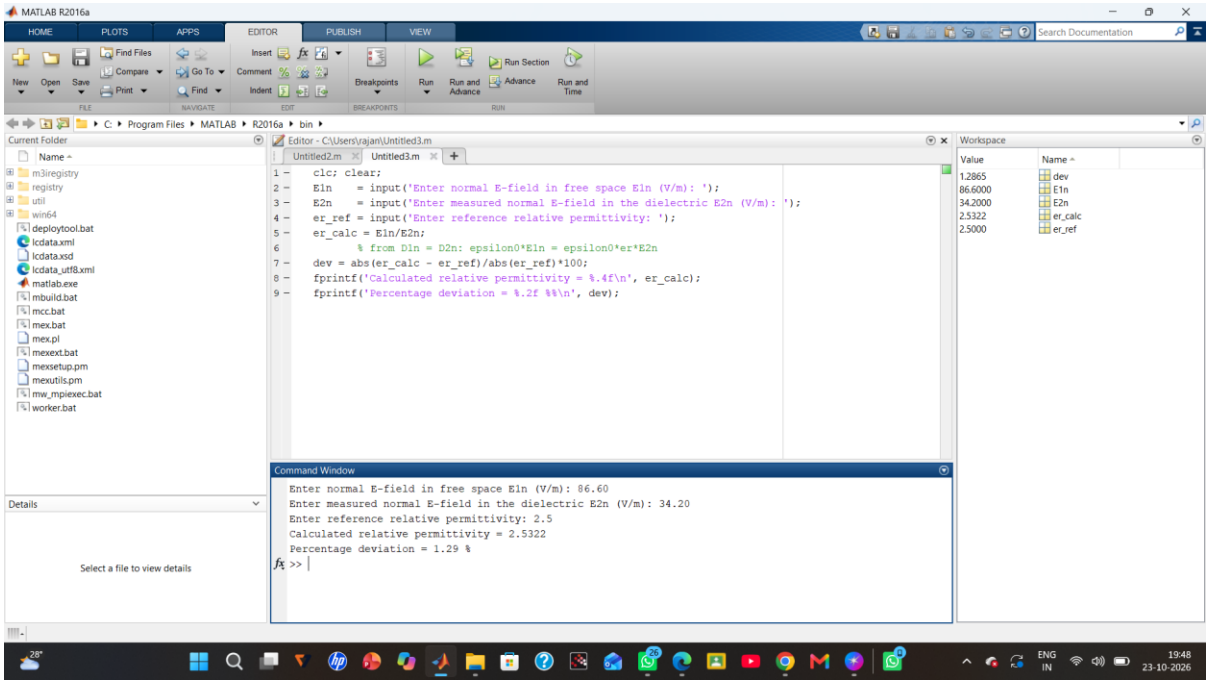
    % from D1n = D2n: epsilon0*E1n = epsilon0*er*E2n
```

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```
dev = abs(er_calc - er_ref)/abs(er_ref)*100;

fprintf('Calculated relative permittivity = %.4f\n', er_calc);

fprintf('Percentage deviation = %.2f %%\n', dev);
```



Test Case (iii) – Variation of the Refraction Angle with the Angle of Incidence for Different Dielectrics

Simulate the interface for three dielectric constants and plot the angle of the electric field in medium 2 against its angle in medium 1.

```
clc; clear;

th1 = linspace(0,89,179);

er  = [1.5 2.5 4];

figure; hold on;

for k = 1:3

th2 = atand(er(k)*tand(th1));      % tan(th2) = er2*tan(th1)

plot(th1,th2,'LineWidth',2);

end

plot(th1,th1,'k--');                % er2 = 1 (no dielectric)

grid on;
```

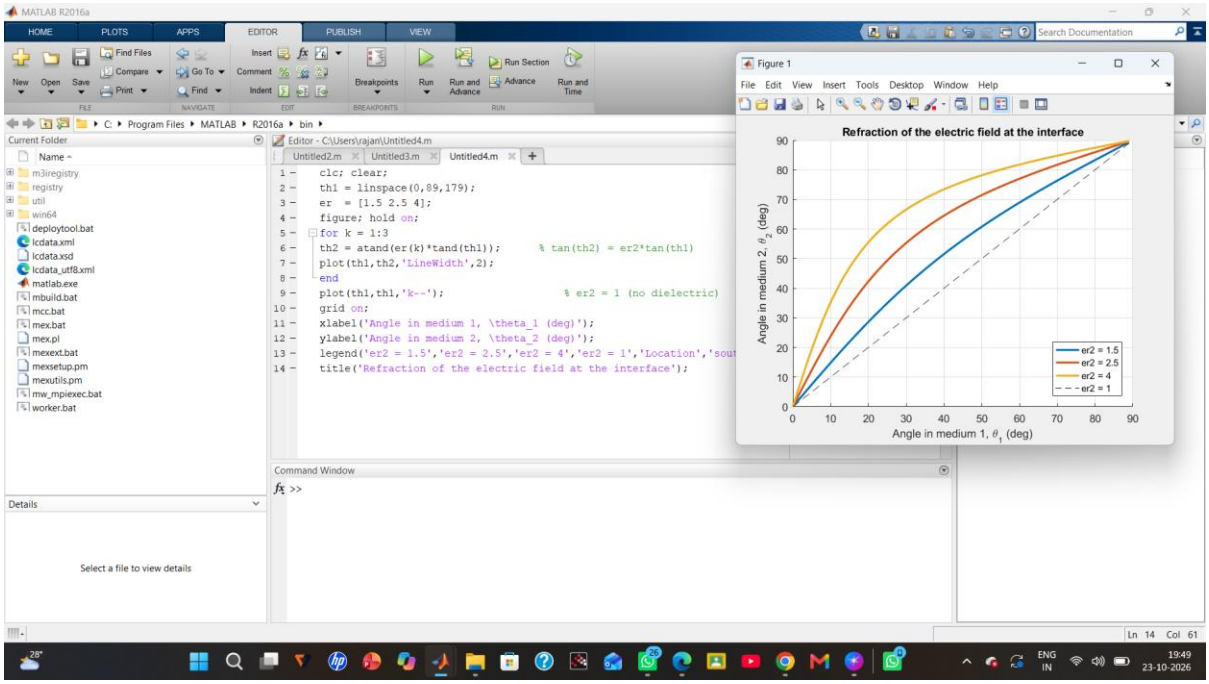
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xlabel('Angle in medium 1, \theta_1 (deg)');

ylabel('Angle in medium 2, \theta_2 (deg)');

legend('er2 = 1.5','er2 = 2.5','er2 = 4','er2 = 1','Location','southeast');

title('Refraction of the electric field at the interface');



Test Case (iv) – Effect of the Relative Permittivity on the Field Components in the Dielectric

Keep the incident field fixed and vary εr2 to show which components are continuous across the boundary.

clc; clear;

epsilon0 = 8.854e-12;

E1 = 100; th1 = 30;

E1n = E1*cosd(th1); E1t = E1*sind(th1);

er2 = linspace(1,6,100);

E2n = E1n./er2;

% normal E in medium 2

E2t = E1t*ones(size(er2)); % tangential E (continuous)

E2 = sqrt(E2n.^2 + E2t.^2);

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```
D2n = epsilon0*er2.*E2n;                % normal D (continuous)

subplot(1,2,1);

plot(er2,E2n,er2,E2t,er2,E2,'LineWidth',2); grid on;

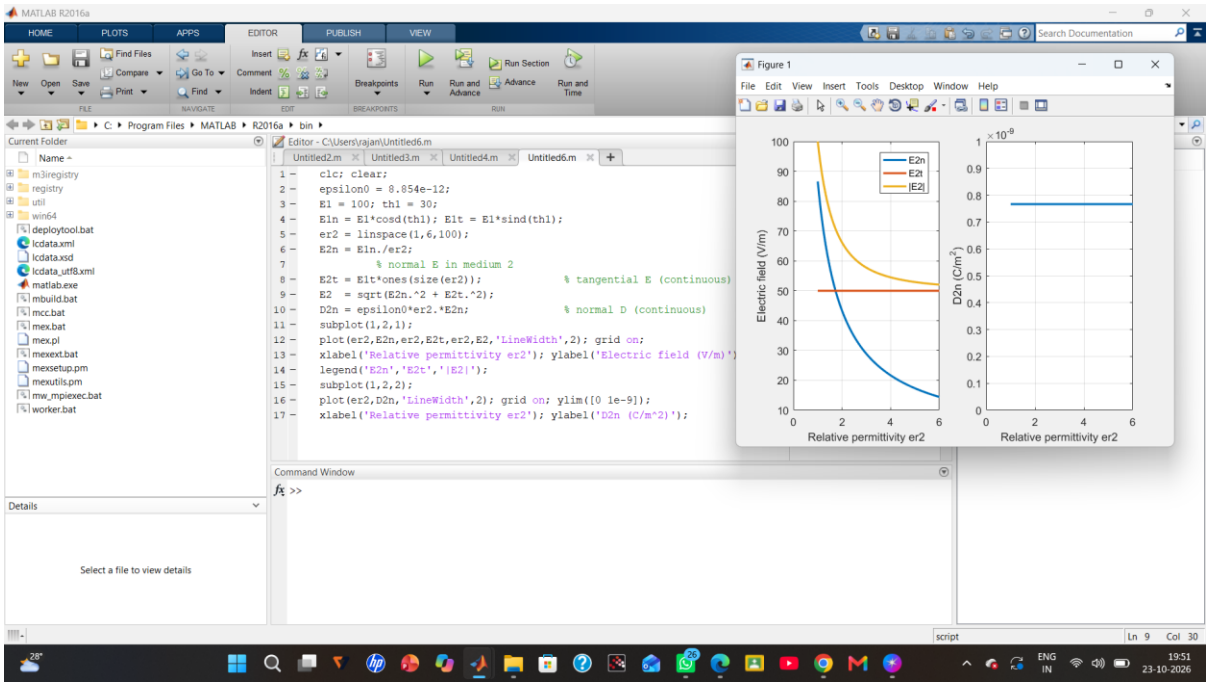
xlabel('Relative permittivity er2'); ylabel('Electric field (V/m)');

legend('E2n','E2t','|E2|');

subplot(1,2,2);

plot(er2,D2n,'LineWidth',2); grid on; ylim([0 1e-9]);

xlabel('Relative permittivity er2'); ylabel('D2n (C/m^2)');
```



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Define interface and field components	Define interface and field components	Correctly identifies media, normal/tangential directions and surface-charge assumptions.	Setup is correct with minor omissions.	Needs guidance on components or assumptions	Interface or component definitions are incorrect.	25	2.5	

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2	Apply electric boundary equations	Apply electric boundary equations	Correctly applies continuity of tangential E and the normal D boundary relation.	Applies relations correctl y with minor errors.	Partly correct application requires support.	Boundary relations are misapplied.	25	2.5	
3	Determine permittivit y and validate	Determine permittivity and validate	Calculates relative permittivity/f ield components consistently and checks boundary conditions.	Results are mostly correct with minor discrepa ncies.	Validation is incomplete or contains errors.	Cannot obtain or validate valid results.	30	3	
4	Interpret and document results	Interpret and document results	Explains field behaviour across the interface and reports units/assump tion s clearly.	Explana tion is correct with minor omissio ns.	Explanation or documentati o n is incomplete.	Results are not interpreted correctly.	20	2	
Tota l							100	10	